

Stage 9: Advanced Search, Filtering & Multi-Faceted Indexing Specification

Executive Overview

Stage 9 provides the high-performance search and filtering engine for the Alight International Multi-Vendor Marketplace. It incorporates PostgreSQL Trigram fuzzy matching, composite query indexing, multi-attribute faceted aggregation, real-time stock availability filtering, and typeahead suggestions.

Key Features & Search Capabilities

1. Multi-Field Fuzzy & Full-Text Search

- **Weighted Target Fields:** Searches product `title`, `short_description`, `description`, `sku`, `tags`, `category.name`, `brand.name`, and `vendor.store_name`.
- **Trigram GIN Indexes (`pg_trgm`):** Sub-string and typo-tolerant search optimization allowing instant search across millions of catalog items.

2. Rich Multi-Faceted Aggregations

- **Category Facets:** Real-time distribution counts across matching categories.
- **Brand Facets:** Dynamic counts of matching brand items.
- **Vendor Facets:** Seller-specific item distribution.
- **Price Brackets:** Dynamic minimum and maximum price range discovery for slider UI rendering.
- **Stock Availability:** Faceted breakdown of `inStockCount` vs `outOfStockCount`.

3. Multi-Faceted Filters & Dynamic Sorting

- **Filters Supported:**
 - `q` : Search keyword or query string - `categoryId` : Category identifier (matches root or children) - `brandId` : Brand identifier - `vendorId` : Specific merchant store - `minPrice` & `maxPrice` : Real-time price boundary - `minRating` : Filter by customer review star threshold - `inStock` : Boolean availability filter (`true` for in-stock, `false` for out-of-stock)
- **Sorting Orders:**
 - `relevance` / default (by review count / popularity) - `price_asc` / `price_low_to_high` - `price_desc` / `price_high_to_low` - `newest` (by `createdAt DESC`) - `rating` / `top_rated` (by `averageRating DESC`)

4. Typeahead Autocomplete Suggestions

- `/api/v1/search/suggestions?q=...` returns instant product previews with title, category, price, and thumbnail image.

Schema Changes (`v35__search_facets_and_indexing.sql`)

1. Trigram & Composite GIN Indexes: - `idx_products_search_composite` on `products(status, category_id, brand_id, base_price)` - `idx_products_title_trgm` on `products` USING `gin (title gin_trgm_ops)` - `idx_products_tags_trgm` on `products` USING `gin (tags gin_trgm_ops)` - `idx_products_sku_trgm` on `products` USING `gin (sku gin_trgm_ops)` - `idx_categories_name_trgm` on `categories` USING `gin (name gin_trgm_ops)` - `idx_brands_name_trgm` on `brands` USING `gin (name gin_trgm_ops)`

API Endpoints (`SearchController.java`)

Method	Endpoint	Query Parameters	Description
GET	<code>/api/v1/search</code>	<code>q</code> , <code>categoryId</code> , <code>brandId</code> , <code>vendorId</code> , <code>minPrice</code> , <code>maxPrice</code> , <code>minRating</code> , <code>inStock</code> , <code>sort</code> , <code>page</code> , <code>size</code>	Full-text search with faceted filters
GET	<code>/api/v1/search/suggestions</code>	<code>q</code> (min 2 chars)	Autocomplete typeahead product suggestions

Integration Tests

- Verified in `ProductSearchIntegrationTest.java`.